Exp No: 11

Date: 28/9/26

a) Stimulate Static Routing configuration using Cisco Packet Tracer.

AIM:

To stimulate and configure static Routing between different network using cisco packet Tracer and to verify communication between the networks using the ping commands.

Procedure:

* Open Cisco Packet Tracer
* place 3 Router, place 3 PCs and connect the PCs to the appropriate Routers.
* Configure each PC with its IP address, Subnet mask and default gateway.

Eg: PC0

IP address: 10.0.0.100

Subnet Mask: 255.0.0.0

Default gateway: 10.0.0.1

* Configure Router Interface using.

Command:

enable

configure terminal

interface gigabitEthernet 0/0

ip address: 198.168.0.1 10.0.00

no shutdown

exit

end.

*. Static routes are manually added using.

Eg. ip route 60.0.0.0 255.0.0.0 40.0.0.2

*. This tells the Router, to reach the 60.0.0.0 network, send the packet to 40.0.0.2.

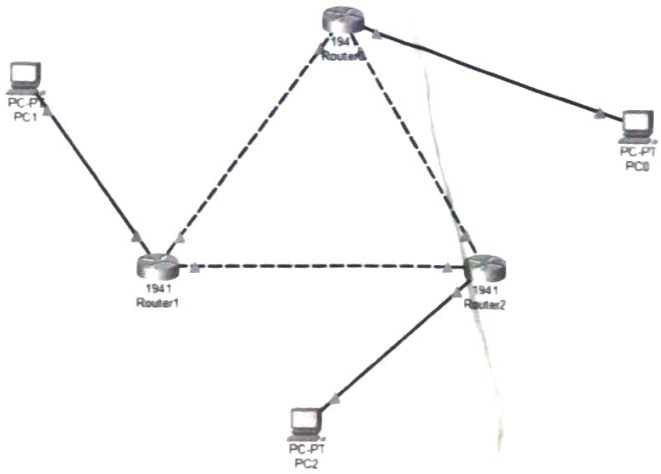
*. A floating static Route can be configured by giving it a higher administrative distance.

*. Test connectivity:

Eg: from PC0:

ping 30.0.0.100.

O/P: Successful replies verify communication between the networks.




```
Router>enable
Router#show ip route static
S       30.0.0.0/8 [20/0] via 40.0.0.2
S       50.0.0.0/8 [10/0] via 40.0.0.2
S       60.0.0.0/8 [1/0] via 40.0.0.2

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#show ip route static
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       50.0.0.0/8 [10/0] via 40.0.0.2
S       60.0.0.0/8 [1/0] via 40.0.0.2
```

The static routing was successfully configured and simulated using Cisco Packet Tracer.

Exp. No: 11
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b) Simulate RIP using CISCO Packet Tracer.

AIM:

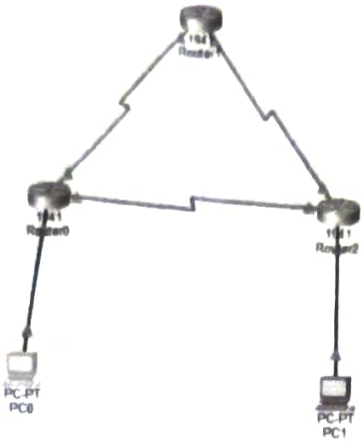
To simulate and configure Routing Information Protocol using Cisco packet Tracer and to Verify dynamic Routing and automatic route selection rules a network path fails.

Procedure:

* Place three Routers & 2 PCs
* Install HWIC-2T Serial modules in the Routers.
* Configure PC0:
  - IP Address: 10.0.0.2
  - Subnet Mask: 255.0.0.0
  - Default gateway: 10.0.0.1
* configure Router0:
  - G0/0 = 10.0.0.1
  - S0/1/0 = 192.168.1.249
  - S0/1/1 = 192.168.1.254
* configure RIP on Router 0.

```
enable
configure terminal
router rip
network 10.0.0.0
network 192.168.1.252
Network 192.168.1.248.
```

* Similarly for all Routers and PCs to configure.
* Test connectivity.




```
Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=58ms TTL=125
Reply from 20.0.0.2: bytes=32 time=56ms TTL=125
Reply from 20.0.0.2: bytes=32 time=60ms TTL=125
Reply from 20.0.0.2: bytes=32 time=55ms TTL=125

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 60ms, Average = 56ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2   11 ms      8 ms      5 ms      192.168.1.250
  3   29 ms     26 ms     25 ms      192.168.1.245
  4   22 ms     28 ms     20 ms      20.0.0.2

Trace complete.
```

Result:

The RIP was successfully configured and simulated using Cisco Packet Tracer.